Evidence-Gated Crop-Loss Claim Verification and Profile-Aware Advisory Chat for Smallholder Farmers: Design, Conformance Evaluation and Live Feasibility Study

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Abstract

Smallholder crop-loss claims are hard to verify, because no single evidence source establishes who may claim or whether a loss occurred. We describe Vivasayi AI, a farmer-facing system for Tamil Nadu, India, that pairs a profile-aware bilingual advisory chat with claim verification in which every evidence source is reduced to an explicit per-component outcome and a deterministic, versioned rule engine, never a language model, decides. We compare \PolicyCountWord{} policies: the historical photograph-only policy; a safe-AI policy, the default as of the change of \PolicySafeAiDate{}, under which neither a model reading alone nor a technical failure can cause an adverse decision; and a required-component gate that makes approval depend on claimant-rights evidence but is not the default, because no authoritative claimant-rights source is available. On a synthetic factorial whose ground truth is the design rule, the photograph-only and safe-AI policies each approved \FusionLegacyInvalidApprovals{} of \FusionInvalidApprovalDenominator{} forbidden scenarios and the gate none; only the photograph-only policy rejected claims on the AI reading alone. In a live feasibility study the production Sentinel-2 step ran on \SatLiveSquares{} parcel-sized areas: near \SatEventCountWord{} documented 2023--2024 flood and cyclone events it reported no supporting loss in any of \SatEventSquares{} areas, and on a harvest control it falsely reported loss in \SatControlSupports{} of \SatControlSquares{}. Live retrieval found a relevant chunk for only \RagAnswerableRelevantHits{} of \RagAnswerableQueries{} answerable questions. Live language-model evaluations were blocked by a free-tier quota. \emph{No result is a measurement on real claims; they show rule enforcement and feasibility, not accuracy.}

Keywords: crop insurance, claim verification, remote sensing, Sentinel-2, NDVI, evidence fusion, retrieval-augmented generation, agricultural advisory, auditability, Tamil

Introduction

Indemnity and index crop insurance depend on knowing whether a loss occurred, on which land, and who may claim it. Remote sensing is used to design area indices, to monitor parcels and to smart-sample crop-cutting experiments [okeyo2020, sitokonstantinou2018, murthy2019smart], and India's YES-TECH programme blends modelled and crop-cutting yields at insurance-unit level [mncfc2023yestech]. Basis risk remains, and resolution and timing affect it [kolle2021, dalhaus2016, carter2017]. Picture-based insurance has been proposed as a farmer-level complement [ceballos2019]. Trust is a barrier to insurance take-up in India [cole2013], and duplicate insurance of the same land has been reported in the press [tribune2023cotton]. At the same time, language and vision-language models can read documents and photographs [gemini2023, kim2022], but hallucination is a documented risk [ji2023], and for high-stakes decisions interpretable rules and audit trails are preferred to opaque models [rudin2019, raji2020].

This paper describes a system built on that premise: models produce evidence; a deterministic engine produces the decision; and every decision is tied to a versioned policy and a snapshot of the evidence it used. Our contributions are:

? A verification design in which each evidence source maps to an explicit per-component outcome, provider failure and missing evidence are kept distinct from negative findings, and versioned policies change how a candidate outcome becomes a decision without changing the frozen rules: a safe-AI policy under which neither a model reading alone nor a technical failure causes an adverse decision, and a required-component

gate that can only make an approval more cautious, holding the claim rather than deciding when evidence is pending.

? An open implementation: a farmer-facing web application and backend with context-sufficiency logic for bilingual chat, parcel-geometry validation, cross-farmer overlap as a review signal, Tamil/Latin name matching, and a pre/post-event Sentinel-2 NDVI step on the claim polygon with scene-classification masking.

? A conformance evaluation on synthetic benchmarks and a live feasibility study (satellite, retrieval, end-to-end farmer journeys), with every post-hoc change disclosed, and a protocol for the real-data experiments that remain.

We do not claim novelty of the individual techniques, which have precedent, nor any accuracy on real claims.

Related Work and Research Gap

Satellite crop-damage assessment. Sentinel-2 [drusch2012] and NDVI [tucker1979] are standard for vegetation monitoring. Pre/post-event NDVI differencing has mapped flood damage to crops, with the authors noting the lack of direct ground validation [ursu2025]; Sentinel-2 has been used for hail [ha2022] and flood damage with in-situ data [bocchino2025], and for land-use anomaly checks in EU subsidy monitoring [kanjir2018]. Closest to our decision logic, a rule-based decision matrix fusing Sentinel-1 and Sentinel-2 evidence mapped jute lodging after Cyclone Amphan in India [chakraborty2021]. Cloud and shadow masking needs validation [baetens2019, raiyani2021, aybar2022], and Tamil Nadu has a public field-level crop dataset [sani2024]. Satellite yield estimation can track ground measures on smallholder plots [lobell2020, burke2017]. Agricultural assistants and grounding. Retrieval-augmented generation [lewis2020, gao2023] has been applied to agriculture with domain constraints [ravindran2026], compared with fine-tuning [balaguer2024], and deployed as personalised farmer chat with LLM-judged faithfulness [singh2024]. Agricultural question-answering benchmarks and systems include AgXQA [kpodo2024] and KisanQRS [rehman2023]; AgriLLM studied Kisan Call Centre queries from Tamil Nadu [didwania2024]. Whether retrieved context is sufficient to answer has been proposed as a lens on retrieval failures [joren2025]; automated RAG metrics exist [es2024]. Voice advisory in India has been evaluated [cole2021], and vision-language models remain weak on fine-grained agricultural recognition [shinoda2025]. Gap. In a search of sources verified from abstracts and bibliographic records plus web documents, extended by sources in a second pass (; the bibliography lists entries, and a source ledger records how much of each was read), we found rule-based fusion of satellite evidence for damage mapping [chakraborty2021] and farmer-level photograph insurance [ceballos2019], but no work that gates individual crop-loss claims on claimant-rights, geometry, photograph, document, satellite and weather components with explicit failure semantics, treats cross-farmer polygon overlap as a review signal, or matches Tamil/Latin names on land documents; nor a public labelled dataset of Indian claim outcomes. Absence in a narrow search is weak evidence, so we make no priority claim.

System Design

The backend is a Node.js service on AWS Lambda with MongoDB; the farmer-facing client is a React web application in English and Tamil. A farmer registers parcels as polygons; a claim is made against a registered parcel with a drawn affected area and a declared relationship to the land. The server computes area, validates geometry, and never accepts area, state or decision from the client.

Evidence components and outcomes

components describe a claim: parcel identity, claimant rights, land documents, geometry, crop, satellite, photographs, weather, overlap and manual review. Each resolves to one of outcomes. A provider that errors or is disabled yields unavailable; missing evidence yields unverified; failed requires a positive finding that

contradicts the claim. This vocabulary is the same under every policy; how a technical failure reaches the decision is policy-specific (Section sec:policies). Evidence derived by a model (photographs, document reading) is never authoritative, and a model reading alone can never verify claimant rights. Satellite and weather are supporting evidence and are never failed. Evidence is append-only: a re-run with new inputs adds a row, and the newest row per component is current.

Deterministic decision and the three policies

A frozen rule engine applies, in order: claim window, event type, area allowance, containment in the parcel, duplicate or partial overlap with the farmer's own claims, remaining eligible area, and the photograph assessment. A versioned policy then decides how that candidate outcome becomes a decision. Each decision stores the policy identifier and a hash of its content, the component outcomes and references to the evidence rows used, so an earlier decision replays under its own policy. The repository defines policies:

? v1-legacy, the historical behaviour and the policy of the live study. Component outcomes are recorded, but none blocks a decision. A confident AI ``no damage'' reading alone rejects the claim. A provider error or timeout leaves the claim undecided and retryable, but an unreadable model answer is treated as an uncertain photograph, and the farmer is asked for more evidence.

? v1.1-safe-ai, the default as of the change of : the legacy rules with four changes. (i) An AI-only ``no damage'' reading holds the claim for human review and opens one review case instead of rejecting it; a rejection by another rule (claim window, event type, area outside the parcel) stands. (ii) A technical failure, including an unreadable model answer, makes the photograph check unavailable: the queue retries it, and the engine holds the claim instead of rejecting it or asking for more evidence. (iii) The claim window is measured at the claim's first submission, not at decision time. (iv) The decision never waits for satellite evidence.

? v2-components, the required-component gate. Parcel identity, geometry, photographs, overlap and claimant rights are required. An approval whose required component is unverified or failed becomes a request for more evidence; one whose required component is pending, unavailable, inconclusive or needs review is held; and a rejection that coincides with a confident AI ``no damage'' reading is held for review. The gate was evaluated but is not the default. A model reading of a document can never verify claimant rights, and no authoritative claimant-rights provider (an authorised land-record service) is available to us, so under the gate no claim could be approved before a reviewer had accepted a land document.

A held claim stays submitted and withdrawable, and no decision is recorded; an administrator may decide it through an audited override. Neither the gate nor a hold ever produces a rejection; rejection stays with the frozen rules and that override. Only under v1-legacy can a model reading alone reject a claim. Under v1-legacy and v2-components an unreadable model answer returns the claim to the farmer for more evidence; under v1.1-safe-ai no technical failure does.

Timing. Slow evidence steps run from a database-backed job queue with leases, so a Lambda that is frozen mid-step cannot lose work. Satellite runs as its own step, independently and including later reschedules while its post-event window is open, and never blocks or changes a decision under any policy: the engine waits for the other queued checks but not for satellite, whose result only adds a supporting evidence row. (In the code of the live study a decision could wait for the post-event window to close.) Under v1.1-safe-ai the claim window is measured at the claim's first submission, which is recorded once and kept on resubmission, so time spent waiting for checks, a resubmission or a review cannot turn into a late-claim rejection. v1-legacy and v2-components still measure it at decision time, so their stored decisions replay unchanged.

Automatic processing. After the study, submission was changed to record the claim's plan of checks durably and idempotently within the request; a worker (a local process and a scheduled Lambda handler) runs due jobs without a browser open, renewing leases and sweeping stuck or lost work; and a farmer-facing tracker shows per-stage progress built only from persisted state, with a satellite status that is never shown as passed unless an analysis supported the loss. These are implemented and covered by unit and integration tests. None

is deployed, and the scheduled worker's trigger is disabled.

Geometry and overlap

Polygons are validated (closed ring, finite in-range coordinates, no self-intersection, non-zero area). Containment uses a tolerance. Overlap with another farmer's claim is found through a bounding-box pre-filter and an exact intersection, and is a review signal only, since shared boundaries, joint holdings and cadastral error are legitimate causes.

Documents and claimant relationship

A vision-language model reads a land document into structured fields; the document is untrusted data, and instruction-like text inside it is itself a finding. Holder names are compared with the claimant across Tamil and Latin scripts by a consonant-skeleton matcher. Each declared relationship (owner, joint owner, tenant, authorised cultivator, heir) has its own expected document set. No authorised Tamil Nadu land-record API is available to us; the official-record adapter is provider-neutral and reports unavailable.

Satellite evidence

For each claim the system requests per-pixel NDVI statistics over the claim polygon from the Copernicus Data Space Statistical API [cdse_statistical] for a pre-event window (event $-$ to $-$ days) and a post-event window ($+$ to $+$ days, capped at the present), masking pixels the Sentinel-2 L2A scene classification marks as no-data, saturated, cloud shadow, cloud or cirrus. Water is deliberately not masked, so flooding stays observable, which is also a confound for paddy. A drop in mean NDVI of at least , with a valid-pixel fraction of at least in both windows, is reported as supporting loss; otherwise no change or inconclusive. The thresholds are provisional and uncalibrated. A change in a vegetation index does not identify a cause. Historical rainfall [openmeteo_hist] is recorded as supporting evidence only. Neither satellite nor weather evidence blocks or changes a decision (Section sec:policies).

Profile-aware advisory chat

For each message the assistant classifies intent, loads only the context that intent needs from the farmer's own records (optionally a selected parcel), tags each item with its provenance (user-provided, verified, third-party, inferred, missing, stale), and asks a follow-up only when a required slot cannot be resolved, about exactly one thing. Retrieved knowledge is fenced and labelled as data, and its source files are returned to the client. Replies follow the requested or detected language.

Evaluation Design

Conformance benchmarks. Deterministic, seeded and reproducible from the repository (npm run eval). Ground truth is analytic (geometry), the design rule (fusion), or a written specification (chat); the name set is author-labelled. No network or real data are used. The v1.1-safe-ai fusion row was computed after the other results, on the same factorial, and written to its own dated result file; the earlier rows still reproduce exactly. Live feasibility study (). (i) Satellite: the production adapter, including the satellite changes made during the study and disclosed in Section sec:disclosures, on ,m squares near documented events, south Tamil Nadu floods of 17--18 December 2023, Cyclone Michaung of 4 December 2023 and Cyclone Fengal of 30 November--1 December 2024, located from humanitarian situation reports, and on a Thanjavur control dated 15 February 2024, within the samba harvest season, with squares each. Land cover, crop and actual damage at each square were not verified, so these are case studies, not labelled data. (ii) Retrieval: queries (answerable from the deployed corpus, advisory) against the production collection, read-only, with relevance by a mechanical rule fixed before the run. (iii) Acceptance: the application's farmer journey (profile, parcel, chat, claim, photographs, land document, submission, verification, cross-user access) run over HTTP against

the real application on a local database with mock storage, mock image AI and mock land records, under v1-legacy and v2-components (the study predates v1.1-safe-ai). (iv) Live chat and synthetic land-document extraction were prepared but could not run (Section sec:blocked).

Results

generated_tables Geometry (Table tab:geometry). Containment, overlap flagging and invalid-geometry detection agree with analytic geometry. The harness exposed two issues: the shared validator accepted out-of-range coordinates (now rejected), and the server's area is about % larger than an ellipsoidal reference at Tamil Nadu latitudes because the formula uses the equatorial radius; we report rather than change it, so stored areas stay comparable. Evidence fusion (Table tab:fusion). The legacy photograph-and-geometry policy approves claims with missing, pending or inconsistent ownership evidence and with overlap on another farmer's claim; the gate does not. v1.1-safe-ai does not reduce invalid approvals relative to v1-legacy: it approves the same of forbidden scenarios, so it guards against adverse decisions, not against wrongful approval. What it changes is the AI-alone rejection: v1-legacy rejected of the scenarios in which only the AI reading stood against the claim, v1.1-safe-ai rejected , and the only decisions that differ between the two policies are rejections produced by the AI rule, which become holds for review. Satellite and weather outcomes never changed a decision in comparisons per policy under any of the policies (under v1.1-safe-ai they did not change a hold either), by design. Because the label is the design rule, this shows rule enforcement, not accuracy. Chat context (Table tab:chat). Against the previous rule, which asked a follow-up whenever a message ended with a question mark, the context logic asked no unnecessary questions and missed none on the specification. The specification and logic share an author, and the baseline is a reimplementation. Satellite, simulated (Table tab:satellite) and live (Table tab:satellitelive). All live runs completed in a median of ,s (maximum ,s). Near the events the rule reported supporting loss in none of squares: showed no change, had too few valid post-event pixels, and returned no valid statistics for the post-event window. of the analysed event squares had a pre-event NDVI between $$ and $$, consistent with bare, flooded or recently transplanted fields, so a fixed window cannot observe a loss there and NDVI rose afterwards. On the harvest control square reported supporting loss ($$NDVI $=$, from to across the harvest), which is a false support produced by normal crop phenology. These observations support the design choice that satellite evidence is optional and never decisive, and they show that a fixed window needs crop-calendar context before thresholds can be calibrated. Retrieval (Table tab:rag). The deployed corpus (chunks from files) consists mostly of district statistics tables and a geography file; it includes a rice pest-prediction table (chunks) whose content we did not inspect, and no retrieved chunk could ground an advisory question (of chunks). For questions the tables can answer, the right file was retrieved for of , but in of such hits the chunk was a header without data rows; a relevant chunk was retrieved for only of . The cause is in ingestion: rows are chunked without the file name or crop, so a district's row is rarely retrieved with its table. Source attribution therefore matters: a farmer and an auditor can see that an answer was not grounded. We do not treat a post-hoc filter replay as a fresh retrieval evaluation: the same historical retrieval set, with header-only chunks removed and raw geography rejected before context assembly, intentionally reduces expected-file hits in the replay without proving an improvement in answerable relevance. The A12 success under the author-defined rule is a file-name match to a pest-prediction table header, not evidence of agronomic advice; under the truth of the corpus it is still not answer-bearing. Acceptance. Under v1-legacy and v2-components every journey step except live chat passed (of under the legacy policy, of under the gated policy), including idempotent replay, cross-user access returning ``not found'', and a client-supplied verification result being ignored. Under the legacy policy the claim was approved on photographs and geometry with claimant rights unverified; under the gated policy it was returned for more evidence, and after a land record was attached and the claim resubmitted it was held with claimant rights pending review, the intended behaviour. No acceptance run under v1.1-safe-ai has been recorded. Name matching. On author-labelled pairs the matcher scored of before a rule change and of after; the change followed those errors on the same set, so the second figure is optimistic.

Disclosures: Defects Found and Post-hoc Changes

The live study found defects that no synthetic test had exposed. We list them because they bear on how far the conformance results can be trusted.

? Response parsing. The production NDVI parser read the provider's statistics at the wrong nesting level, so every live analysis in our first runs returned ``no valid statistics'' (those runs were not saved). It was fixed, and a recorded live response is now a regression fixture.

? Scene cloud filter (post-hoc). A % scene-level cloud limit rejected all monsoon imagery for the event windows in our first live runs (not saved; recorded only in a code comment). We raised the scene limit used for parcel statistics to % after observing this, relying on per-pixel masking for the parcel itself. The NDVI thresholds were not changed.

? Prototype request. The earlier proof-of-concept request was rejected by the live service (HTTP , missing data-mask output); it had never run live. Production no longer uses it.

? Quota masked as timeout. A model-provider quota refusal was retried with back-off until the ,s timeout fired, so farmers would have seen ``did not respond in time''. Retries are now capped and the refusal is reported as a temporary unavailability.

? Stale decision on hold. A claim held after a resubmission still reported the previous round's decision. The hold now clears it; the earlier decision remains on the audit trail.

? Write budget. A realistic single-claim journey exceeds the per-user limit of claim writes per hour in force during the study; that value is kept in a pinned record of the study-time code defaults. After the study the owner raised the defaults to claim writes and evidence requests per user per hour (); this change is not deployed, and no study result depends on it.

Blocked Evaluations

The configured language-model key was on the provider's free tier, and its daily request quota for the model used was exhausted during the study (the quota was diagnosed from the provider's error during the study and is not recorded in a saved result). The live chat comparison (context-aware versus generic, items) and synthetic land-document extraction (rendered, watermarked documents, with injected instructions) are implemented and ready to run but produced no results, and we report none. Latency, token and cost figures for chat are therefore absent.

Discussion

The structural finding is that separating evidence from decision, and failure from negative finding, makes behaviour under outages and missing data explicit and testable, and lets a stricter or safer policy be introduced without changing the frozen rules or the replay of earlier decisions. The safe-AI policy shows the limit of that separation as well as its use: it removes adverse decisions caused by a model reading or an outage, but it does not change which claims are approved. The live study adds two empirical cautions that the synthetic study could not: a fixed-window NDVI change rule is confounded by crop phenology and sparse post-event observation, and retrieval quality depends on corpus content and chunking more than on the retriever. Both favour the conservative roles the design assigns to these sources. The retrieval lesson is also operational: a header-only filter improves evidence hygiene and blocks malformed context, but it is not a legitimate retrieval-quality improvement when the same historical retrieval set is replayed after filtering; a fresh evaluation would require a new corpus and a new baseline, which we do not claim.

Limitations and Threats to Validity

? No real claims, farmers or documents were used. No accuracy, fraud-detection or loss-estimation figure is claimed.

? Live satellite cases are squares chosen near documented events without verifying land cover, crop or damage; they are not labelled data, and one parameter was changed after observation.

? Specifications and relevance rules are author-defined; the name matcher was tuned on its own evaluation set.

? v1.1-safe-ai is evaluated only on the synthetic factorial and by automated tests; no live or acceptance run used it, and technical failures are not a factor of the factorial.

? Live language-model behaviour is unmeasured. The deployed knowledge base holds almost no advisory text (one rice pest-prediction table, not inspected), and the original retrieval baseline remains poor even after header-only chunks are filtered.

? The retrieval replay is a post-hoc safety check, not a fresh evaluation: changing the context assembly after the fact does not create a new evidence base or improved relevance labels.

? No authorised land-record API is available; the required-document policy per relationship is unreviewed by any legal or insurer authority.

Ethical, Legal and Privacy Considerations

No real farmer data were used; the synthetic documents are watermarked and use fictional names. Real use needs informed consent, secure handling of documents, permission for any official record and human review of adverse and uncertain outcomes; under v1.1-safe-ai and v2-components no model reading alone rejects a claim, while under v1-legacy one can. The statement in the repository has not been reviewed by a lawyer or ethics board.

Conclusion and Future Work

We described, implemented and tested an evidence-gated claim-verification design with versioned policies, including a safe-AI default, and a profile-aware advisory chat, and ran the satellite, retrieval and farmer-journey paths live. The results show that the components enforce their specifications and that the live paths work; they do not show real-world accuracy. Next steps are: live chat answers rated by agronomists in English and Tamil; labelled document extraction; satellite validation against independently labelled parcels with crop-calendar-aware windows and Sentinel-1 for cloudy periods; re-chunking the knowledge base with advisory sources; and evaluation on authorised cadastral data. IEEEtran ../references